

15.4 Double Integrals in Polar Form

Integrals are sometimes easier to evaluate if we change to polar coordinates. This section shows how to accomplish the change and how to evaluate integrals over regions whose boundaries are given by polar equations.

Integrals in Polar Coordinates

When we defined the double integral of a function over a region R in the xy -plane, we began by cutting R into rectangles whose sides were parallel to the coordinate axes. These were the natural shapes to use because their sides have either constant x -values or constant y -values. In polar coordinates, the natural shape is a “polar rectangle” whose sides have constant r - and θ -values.

Suppose that a function $f(r, \theta)$ is defined over a region R that is bounded by the rays $\theta = \alpha$ and $\theta = \beta$ and by the continuous curves $r = g_1(\theta)$ and $r = g_2(\theta)$. Suppose also that $0 \leq g_1(\theta) \leq g_2(\theta) \leq a$ for every value of θ between α and β . Then R lies in a fan-shaped region Q defined by the inequalities $0 \leq r \leq a$ and $\alpha \leq \theta \leq \beta$. See Figure 15.21.

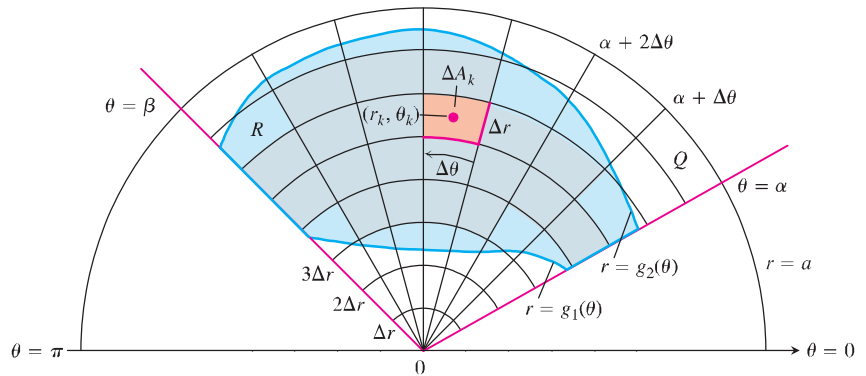


FIGURE 15.21 The region $R: g_1(\theta) \leq r \leq g_2(\theta), \alpha \leq \theta \leq \beta$, is contained in the fan-shaped region $Q: 0 \leq r \leq a, \alpha \leq \theta \leq \beta$. The partition of Q by circular arcs and rays induces a partition of R .

We cover Q by a grid of circular arcs and rays. The arcs are cut from circles centered at the origin, with radii $\Delta r, 2\Delta r, \dots, m\Delta r$, where $\Delta r = a/m$. The rays are given by

$$\theta = \alpha, \quad \theta = \alpha + \Delta\theta, \quad \theta = \alpha + 2\Delta\theta, \quad \dots, \quad \theta = \alpha + m'\Delta\theta = \beta,$$

where $\Delta\theta = (\beta - \alpha)/m'$. The arcs and rays partition Q into small patches called “polar rectangles.”

We number the polar rectangles that lie inside R (the order does not matter), calling their areas $\Delta A_1, \Delta A_2, \dots, \Delta A_n$. We let (r_k, θ_k) be any point in the polar rectangle whose area is ΔA_k . We then form the sum

$$S_n = \sum_{k=1}^n f(r_k, \theta_k) \Delta A_k.$$

If f is continuous throughout R , this sum will approach a limit as we refine the grid to make Δr and $\Delta\theta$ go to zero. The limit is called the double integral of f over R . In symbols,

$$\lim_{n \rightarrow \infty} S_n = \iint_R f(r, \theta) dA.$$

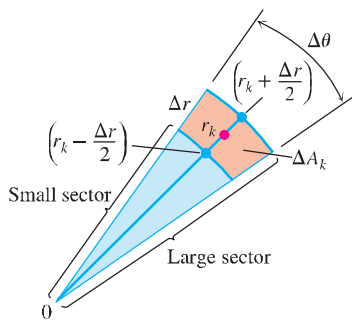


FIGURE 15.22 The observation that

$$\Delta A_k = \left(\text{area of large sector} \right) - \left(\text{area of small sector} \right)$$

leads to the formula $\Delta A_k = r_k \Delta r \Delta \theta$.

To evaluate this limit, we first have to write the sum S_n in a way that expresses ΔA_k in terms of Δr and $\Delta \theta$. For convenience we choose r_k to be the average of the radii of the inner and outer arcs bounding the k th polar rectangle ΔA_k . The radius of the inner arc bounding ΔA_k is then $r_k - (\Delta r/2)$ (Figure 15.22). The radius of the outer arc is $r_k + (\Delta r/2)$.

The area of a wedge-shaped sector of a circle having radius r and angle θ is

$$A = \frac{1}{2} \theta \cdot r^2,$$

as can be seen by multiplying πr^2 , the area of the circle, by $\theta/2\pi$, the fraction of the circle's area contained in the wedge. So the areas of the circular sectors subtended by these arcs at the origin are

$$\text{Inner radius: } \frac{1}{2} \left(r_k - \frac{\Delta r}{2} \right)^2 \Delta \theta$$

$$\text{Outer radius: } \frac{1}{2} \left(r_k + \frac{\Delta r}{2} \right)^2 \Delta \theta.$$

Therefore,

$$\begin{aligned} \Delta A_k &= \text{area of large sector} - \text{area of small sector} \\ &= \frac{\Delta \theta}{2} \left[\left(r_k + \frac{\Delta r}{2} \right)^2 - \left(r_k - \frac{\Delta r}{2} \right)^2 \right] = \frac{\Delta \theta}{2} (2r_k \Delta r) = r_k \Delta r \Delta \theta. \end{aligned}$$

Combining this result with the sum defining S_n gives

$$S_n = \sum_{k=1}^n f(r_k, \theta_k) r_k \Delta r \Delta \theta.$$

As $n \rightarrow \infty$ and the values of Δr and $\Delta \theta$ approach zero, these sums converge to the double integral

$$\lim_{n \rightarrow \infty} S_n = \iint_R f(r, \theta) r \, dr \, d\theta.$$

A version of Fubini's Theorem says that the limit approached by these sums can be evaluated by repeated single integrations with respect to r and θ as

$$\iint_R f(r, \theta) \, dA = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=g_1(\theta)}^{r=g_2(\theta)} f(r, \theta) r \, dr \, d\theta.$$

Finding Limits of Integration

The procedure for finding limits of integration in rectangular coordinates also works for polar coordinates. To evaluate $\iint_R f(r, \theta) \, dA$ over a region R in polar coordinates, integrating first with respect to r and then with respect to θ , take the following steps.

1. *Sketch.* Sketch the region and label the bounding curves (Figure 15.23a).
2. *Find the r -limits of integration.* Imagine a ray L from the origin cutting through R in the direction of increasing r . Mark the r -values where L enters and leaves R . These are the r -limits of integration. They usually depend on the angle θ that L makes with the positive x -axis (Figure 15.23b).
3. *Find the θ -limits of integration.* Find the smallest and largest θ -values that bound R . These are the θ -limits of integration (Figure 15.23c). The polar iterated integral is

$$\iint_R f(r, \theta) \, dA = \int_{\theta=\pi/4}^{\theta=\pi/2} \int_{r=\sqrt{2} \csc \theta}^{r=2} f(r, \theta) r \, dr \, d\theta.$$

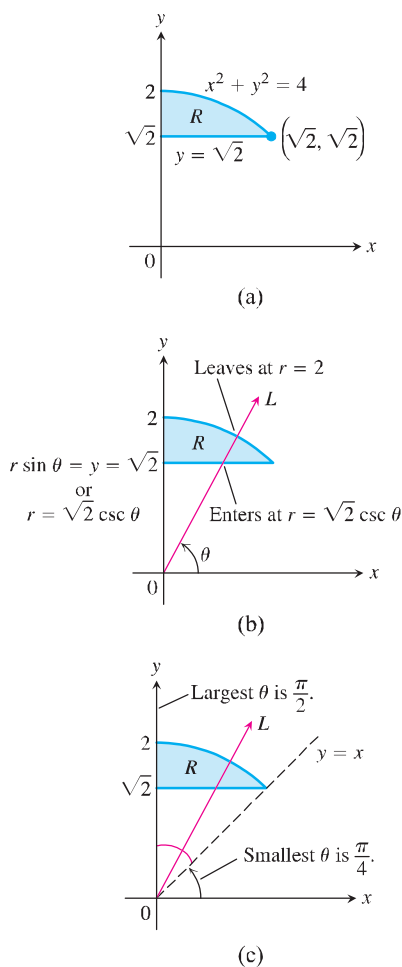


FIGURE 15.23 Finding the limits of integration in polar coordinates.

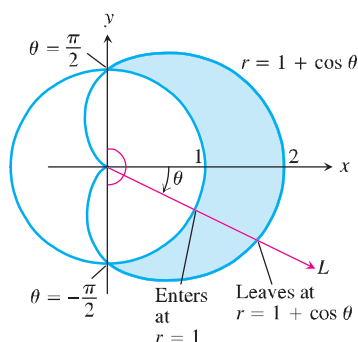


FIGURE 15.24 Finding the limits of integration in polar coordinates for the region in Example 1.

Area Differential in Polar Coordinates

$$dA = r \, dr \, d\theta$$

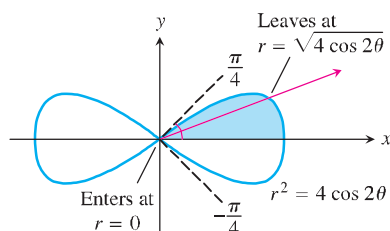


FIGURE 15.25 To integrate over the shaded region, we run r from 0 to $\sqrt{4 \cos 2\theta}$ and θ from 0 to $\pi/4$ (Example 2).

EXAMPLE 1 Find the limits of integration for integrating $f(r, \theta)$ over the region R that lies inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 1$.

Solution

1. We first sketch the region and label the bounding curves (Figure 15.24).
2. Next we find the r -limits of integration. A typical ray from the origin enters R where $r = 1$ and leaves where $r = 1 + \cos \theta$.
3. Finally we find the θ -limits of integration. The rays from the origin that intersect R run from $\theta = -\pi/2$ to $\theta = \pi/2$. The integral is

$$\int_{-\pi/2}^{\pi/2} \int_1^{1+\cos \theta} f(r, \theta) \, r \, dr \, d\theta.$$

If $f(r, \theta)$ is the constant function whose value is 1, then the integral of f over R is the area of R .

Area in Polar Coordinates

The area of a closed and bounded region R in the polar coordinate plane is

$$A = \iint_R r \, dr \, d\theta.$$

This formula for area is consistent with all earlier formulas, although we do not prove this fact.

EXAMPLE 2 Find the area enclosed by the lemniscate $r^2 = 4 \cos 2\theta$.

Solution We graph the lemniscate to determine the limits of integration (Figure 15.25) and see from the symmetry of the region that the total area is 4 times the first-quadrant portion.

$$\begin{aligned} A &= 4 \int_0^{\pi/4} \int_0^{\sqrt{4 \cos 2\theta}} r \, dr \, d\theta = 4 \int_0^{\pi/4} \left[\frac{r^2}{2} \right]_{r=0}^{r=\sqrt{4 \cos 2\theta}} d\theta \\ &= 4 \int_0^{\pi/4} 2 \cos 2\theta \, d\theta = 4 \sin 2\theta \Big|_0^{\pi/4} = 4. \end{aligned}$$

Changing Cartesian Integrals into Polar Integrals

The procedure for changing a Cartesian integral $\iint_R f(x, y) \, dx \, dy$ into a polar integral has two steps. First substitute $x = r \cos \theta$ and $y = r \sin \theta$, and replace $dx \, dy$ by $r \, dr \, d\theta$ in the Cartesian integral. Then supply polar limits of integration for the boundary of R . The Cartesian integral then becomes

$$\iint_R f(x, y) \, dx \, dy = \iint_G f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta,$$

where G denotes the same region of integration now described in polar coordinates. This is like the substitution method in Chapter 5 except that there are now two variables to substitute for instead of one. Notice that the area differential $dx \, dy$ is not replaced by $dr \, d\theta$ but by $r \, dr \, d\theta$. A more general discussion of changes of variables (substitutions) in multiple integrals is given in Section 15.8.

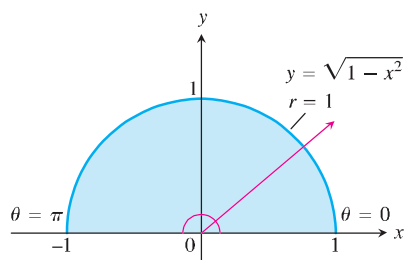


FIGURE 15.26 The semicircular region in Example 3 is the region

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq \pi.$$

EXAMPLE 3 Evaluate

$$\iint_R e^{x^2+y^2} dy dx,$$

where R is the semicircular region bounded by the x -axis and the curve $y = \sqrt{1-x^2}$ (Figure 15.26).

Solution In Cartesian coordinates, the integral in question is a nonelementary integral and there is no direct way to integrate $e^{x^2+y^2}$ with respect to either x or y . Yet this integral and others like it are important in mathematics—in statistics, for example—and we need to find a way to evaluate it. Polar coordinates save the day. Substituting $x = r \cos \theta$, $y = r \sin \theta$ and replacing $dy dx$ by $r dr d\theta$ enables us to evaluate the integral as

$$\begin{aligned} \iint_R e^{x^2+y^2} dy dx &= \int_0^\pi \int_0^1 e^{r^2} r dr d\theta = \int_0^\pi \left[\frac{1}{2} e^{r^2} \right]_0^1 d\theta \\ &= \int_0^\pi \frac{1}{2} (e - 1) d\theta = \frac{\pi}{2} (e - 1). \end{aligned}$$

The r in the $r dr d\theta$ was just what we needed to integrate e^{r^2} . Without it, we would have been unable to find an antiderivative for the first (innermost) iterated integral. ■

EXAMPLE 4 Evaluate the integral

$$\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx.$$

Solution Integration with respect to y gives

$$\int_0^1 \left(x^2 \sqrt{1-x^2} + \frac{(1-x^2)^{3/2}}{3} \right) dx,$$

an integral difficult to evaluate without tables.

Things go better if we change the original integral to polar coordinates. The region of integration in Cartesian coordinates is given by the inequalities $0 \leq y \leq \sqrt{1-x^2}$ and $0 \leq x \leq 1$, which correspond to the interior of the unit quarter circle $x^2 + y^2 = 1$ in the first quadrant. (See Figure 15.26, first quadrant.) Substituting the polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, $0 \leq \theta \leq \pi/2$ and $0 \leq r \leq 1$, and replacing $dx dy$ by $r dr d\theta$ in the double integral, we get

$$\begin{aligned} \int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx &= \int_0^{\pi/2} \int_0^1 (r^2) r dr d\theta \\ &= \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_{r=0}^{r=1} d\theta = \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{\pi}{8}. \end{aligned}$$

Why is the polar coordinate transformation so effective here? One reason is that $x^2 + y^2$ simplifies to r^2 . Another is that the limits of integration become constants. ■

EXAMPLE 5 Find the volume of the solid region bounded above by the paraboloid $z = 9 - x^2 - y^2$ and below by the unit circle in the xy -plane.

Solution The region of integration R is the unit circle $x^2 + y^2 = 1$, which is described in polar coordinates by $r = 1$, $0 \leq \theta \leq 2\pi$. The solid region is shown in Figure 15.27. The volume is given by the double integral

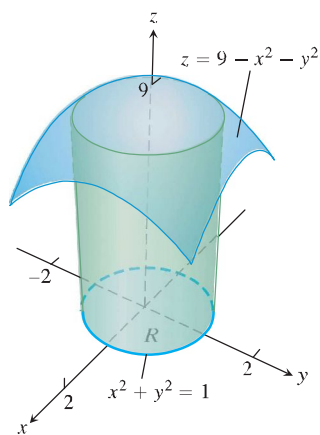


FIGURE 15.27 The solid region in Example 5.

$$\begin{aligned}
 \iint_R (9 - x^2 - y^2) dA &= \int_0^{2\pi} \int_0^1 (9 - r^2) r dr d\theta \\
 &= \int_0^{2\pi} \int_0^1 (9r - r^3) dr d\theta \\
 &= \int_0^{2\pi} \left[\frac{9}{2} r^2 - \frac{1}{4} r^4 \right]_{r=0}^{r=1} d\theta \\
 &= \frac{17}{4} \int_0^{2\pi} d\theta = \frac{17\pi}{2}.
 \end{aligned}$$

EXAMPLE 6 Using polar integration, find the area of the region R in the xy -plane enclosed by the circle $x^2 + y^2 = 4$, above the line $y = 1$, and below the line $y = \sqrt{3}x$.

Solution A sketch of the region R is shown in Figure 15.28. First we note that the line $y = \sqrt{3}x$ has slope $\sqrt{3} = \tan \theta$, so $\theta = \pi/3$. Next we observe that the line $y = 1$ intersects the circle $x^2 + y^2 = 4$ when $x^2 + 1 = 4$, or $x = \sqrt{3}$. Moreover, the radial line from the origin through the point $(\sqrt{3}, 1)$ has slope $1/\sqrt{3} = \tan \theta$, giving its angle of inclination as $\theta = \pi/6$. This information is shown in Figure 15.28.

Now, for the region R , as θ varies from $\pi/6$ to $\pi/3$, the polar coordinate r varies from the horizontal line $y = 1$ to the circle $x^2 + y^2 = 4$. Substituting $r \sin \theta$ for y in the equation for the horizontal line, we have $r \sin \theta = 1$, or $r = \csc \theta$, which is the polar equation of the line. The polar equation for the circle is $r = 2$. So in polar coordinates, for $\pi/6 \leq \theta \leq \pi/3$, r varies from $r = \csc \theta$ to $r = 2$. It follows that the iterated integral for the area then gives

$$\begin{aligned}
 \iint_R dA &= \int_{\pi/6}^{\pi/3} \int_{\csc \theta}^2 r dr d\theta \\
 &= \int_{\pi/6}^{\pi/3} \left[\frac{1}{2} r^2 \right]_{r=\csc \theta}^{r=2} d\theta \\
 &= \int_{\pi/6}^{\pi/3} \frac{1}{2} [4 - \csc^2 \theta] d\theta \\
 &= \frac{1}{2} [4\theta + \cot \theta]_{\pi/6}^{\pi/3} \\
 &= \frac{1}{2} \left(\frac{4\pi}{3} + \frac{1}{\sqrt{3}} \right) - \frac{1}{2} \left(\frac{4\pi}{6} + \sqrt{3} \right) = \frac{\pi - \sqrt{3}}{3}.
 \end{aligned}$$

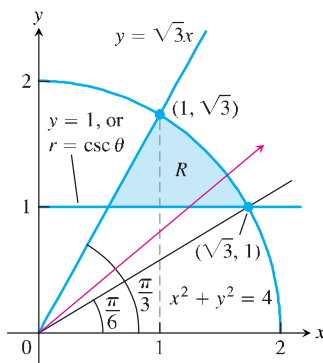


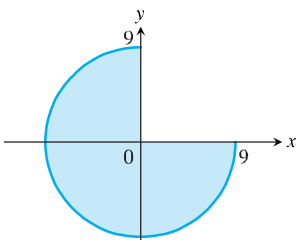
FIGURE 15.28 The region R in Example 6.

Exercises 15.4

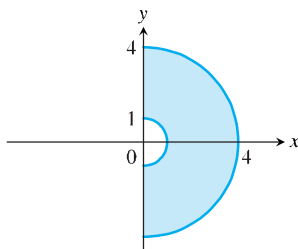
Regions in Polar Coordinates

In Exercises 1–8, describe the given region in polar coordinates.

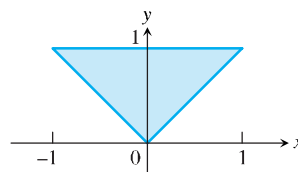
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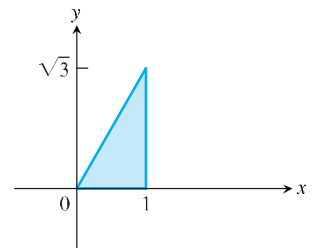
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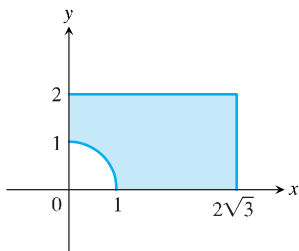
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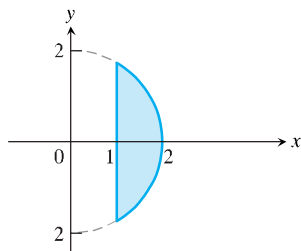
4.



5.



6.



7. The region enclosed by the circle $x^2 + y^2 = 2x$.
 8. The region enclosed by the semicircle $x^2 + y^2 = 2y, y \geq 0$.

Evaluating Polar Integrals

In Exercises 9–22, change the Cartesian integral into an equivalent polar integral. Then evaluate the polar integral.

9. $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} dy \, dx$
10. $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) \, dx \, dy$
11. $\int_0^2 \int_0^{\sqrt{4-y^2}} (x^2 + y^2) \, dx \, dy$
12. $\int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dy \, dx$
13. $\int_0^6 \int_0^y x \, dx \, dy$
14. $\int_0^2 \int_0^x y \, dy \, dx$
15. $\int_1^{\sqrt{3}} \int_1^x dy \, dx$
16. $\int_{\sqrt{2}}^2 \int_{\sqrt{4-y^2}}^y dx \, dy$
17. $\int_{-1}^0 \int_{-\sqrt{1-x^2}}^0 \frac{2}{1 + \sqrt{x^2 + y^2}} dy \, dx$
18. $\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{(1 + x^2 + y^2)^2} dy \, dx$
19. $\int_0^{\ln 2} \int_0^{\sqrt{(\ln 2)^2 - y^2}} e^{\sqrt{x^2 + y^2}} dx \, dy$
20. $\int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \ln(x^2 + y^2 + 1) \, dx \, dy$
21. $\int_0^1 \int_x^{\sqrt{2-x^2}} (x + 2y) \, dy \, dx$
22. $\int_1^2 \int_0^{\sqrt{2x-x^2}} \frac{1}{(x^2 + y^2)^2} dy \, dx$

In Exercises 23–26, sketch the region of integration and convert each polar integral or sum of integrals to a Cartesian integral or sum of integrals. Do not evaluate the integrals.

23. $\int_0^{\pi/2} \int_0^1 r^3 \sin \theta \cos \theta \, dr \, d\theta$
24. $\int_{\pi/6}^{\pi/2} \int_1^{\csc \theta} r^2 \cos \theta \, dr \, d\theta$
25. $\int_0^{\pi/4} \int_0^{2 \sec \theta} r^5 \sin^2 \theta \, dr \, d\theta$
26. $\int_0^{\tan^{-1} \frac{4}{3}} \int_0^{3 \sec \theta} r^7 \, dr \, d\theta + \int_{\tan^{-1} \frac{4}{3}}^{\pi/2} \int_0^{4 \csc \theta} r^7 \, dr \, d\theta$

Area in Polar Coordinates

27. Find the area of the region cut from the first quadrant by the curve $r = 2(2 - \sin 2\theta)^{1/2}$.
28. **Cardioid overlapping a circle** Find the area of the region that lies inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 1$.
29. **One leaf of a rose** Find the area enclosed by one leaf of the rose $r = 12 \cos 3\theta$.
30. **Snail shell** Find the area of the region enclosed by the positive x -axis and spiral $r = 4\theta/3, 0 \leq \theta \leq 2\pi$. The region looks like a snail shell.
31. **Cardioid in the first quadrant** Find the area of the region cut from the first quadrant by the cardioid $r = 1 + \sin \theta$.
32. **Overlapping cardioids** Find the area of the region common to the interiors of the cardioids $r = 1 + \cos \theta$ and $r = 1 - \cos \theta$.

Average values

In polar coordinates, the **average value** of a function over a region R (Section 15.3) is given by

$$\frac{1}{\text{Area}(R)} \iint_R f(r, \theta) r \, dr \, d\theta.$$

33. **Average height of a hemisphere** Find the average height of the hemispherical surface $z = \sqrt{a^2 - x^2 - y^2}$ above the disk $x^2 + y^2 \leq a^2$ in the xy -plane.
34. **Average height of a cone** Find the average height of the (single) cone $z = \sqrt{x^2 + y^2}$ above the disk $x^2 + y^2 \leq a^2$ in the xy -plane.
35. **Average distance from interior of disk to center** Find the average distance from a point $P(x, y)$ in the disk $x^2 + y^2 \leq a^2$ to the origin.
36. **Average distance squared from a point in a disk to a point in its boundary** Find the average value of the *square* of the distance from the point $P(x, y)$ in the disk $x^2 + y^2 \leq 1$ to the boundary point $A(1, 0)$.

Theory and Examples

37. **Converting to a polar integral** Integrate $f(x, y) = [\ln(x^2 + y^2)]/\sqrt{x^2 + y^2}$ over the region $1 \leq x^2 + y^2 \leq e$.
38. **Converting to a polar integral** Integrate $f(x, y) = [\ln(x^2 + y^2)]/(x^2 + y^2)$ over the region $1 \leq x^2 + y^2 \leq e^2$.
39. **Volume of noncircular right cylinder** The region that lies inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 1$ is the base of a solid right cylinder. The top of the cylinder lies in the plane $z = x$. Find the cylinder's volume.
40. **Volume of noncircular right cylinder** The region enclosed by the lemniscate $r^2 = 2 \cos 2\theta$ is the base of a solid right cylinder whose top is bounded by the sphere $z = \sqrt{2 - r^2}$. Find the cylinder's volume.
41. **Converting to polar integrals**
 - a. The usual way to evaluate the improper integral $I = \int_0^\infty e^{-x^2} dx$ is first to calculate its square:
$$I^2 = \left(\int_0^\infty e^{-x^2} dx \right) \left(\int_0^\infty e^{-y^2} dy \right) = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx \, dy.$$

Evaluate the last integral using polar coordinates and solve the resulting equation for I .

b. Evaluate

$$\lim_{x \rightarrow \infty} \operatorname{erf}(x) = \lim_{x \rightarrow \infty} \int_0^x \frac{2e^{-t^2}}{\sqrt{\pi}} dt.$$

42. **Converting to a polar integral** Evaluate the integral

$$\int_0^\infty \int_0^\infty \frac{1}{(1+x^2+y^2)^2} dx dy.$$

43. **Existence** Integrate the function $f(x, y) = 1/(1 - x^2 - y^2)$ over the disk $x^2 + y^2 \leq 3/4$. Does the integral of $f(x, y)$ over the disk $x^2 + y^2 \leq 1$ exist? Give reasons for your answer.

44. **Area formula in polar coordinates** Use the double integral in polar coordinates to derive the formula

$$A = \int_\alpha^\beta \frac{1}{2} r^2 d\theta$$

for the area of the fan-shaped region between the origin and polar curve $r = f(\theta)$, $\alpha \leq \theta \leq \beta$.

45. **Average distance to a given point inside a disk** Let P_0 be a point inside a circle of radius a and let h denote the distance from P_0 to the center of the circle. Let d denote the distance from an arbitrary point P to P_0 . Find the average value of d^2 over the region enclosed by the circle. (Hint: Simplify your work by placing the center of the circle at the origin and P_0 on the x -axis.)

46. **Area** Suppose that the area of a region in the polar coordinate plane is

$$A = \int_{\pi/4}^{3\pi/4} \int_{\csc \theta}^{2 \sin \theta} r dr d\theta.$$

Sketch the region and find its area.

COMPUTER EXPLORATIONS

In Exercises 47–50, use a CAS to change the Cartesian integrals into an equivalent polar integral and evaluate the polar integral. Perform the following steps in each exercise.

- Plot the Cartesian region of integration in the xy -plane.
- Change each boundary curve of the Cartesian region in part (a) to its polar representation by solving its Cartesian equation for r and θ .
- Using the results in part (b), plot the polar region of integration in the $r\theta$ -plane.
- Change the integrand from Cartesian to polar coordinates. Determine the limits of integration from your plot in part (c) and evaluate the polar integral using the CAS integration utility.

$$47. \int_0^1 \int_x^1 \frac{y}{x^2 + y^2} dy dx \quad 48. \int_0^1 \int_0^{x/2} \frac{x}{x^2 + y^2} dy dx$$

$$49. \int_0^1 \int_{-y/3}^{y/3} \frac{y}{\sqrt{x^2 + y^2}} dx dy \quad 50. \int_0^1 \int_y^{2-y} \sqrt{x + y} dx dy$$

15.5 Triple Integrals in Rectangular Coordinates

Just as double integrals allow us to deal with more general situations than could be handled by single integrals, triple integrals enable us to solve still more general problems. We use triple integrals to calculate the volumes of three-dimensional shapes and the average value of a function over a three-dimensional region. Triple integrals also arise in the study of vector fields and fluid flow in three dimensions, as we will see in Chapter 16.

Triple Integrals

If $F(x, y, z)$ is a function defined on a closed, bounded region D in space, such as the region occupied by a solid ball or a lump of clay, then the integral of F over D may be defined in the following way. We partition a rectangular boxlike region containing D into rectangular cells by planes parallel to the coordinate axes (Figure 15.29). We number the cells that lie completely inside D from 1 to n in some order, the k th cell having dimensions Δx_k by Δy_k by Δz_k and volume $\Delta V_k = \Delta x_k \Delta y_k \Delta z_k$. We choose a point (x_k, y_k, z_k) in each cell and form the sum

$$S_n = \sum_{k=1}^n F(x_k, y_k, z_k) \Delta V_k. \quad (1)$$

We are interested in what happens as D is partitioned by smaller and smaller cells, so that Δx_k , Δy_k , Δz_k and the norm of the partition $\|P\|$, the largest value among Δx_k , Δy_k , Δz_k , all approach zero. When a single limiting value is attained, no matter how the partitions and points (x_k, y_k, z_k) are chosen, we say that F is **integrable** over D . As before, it can be

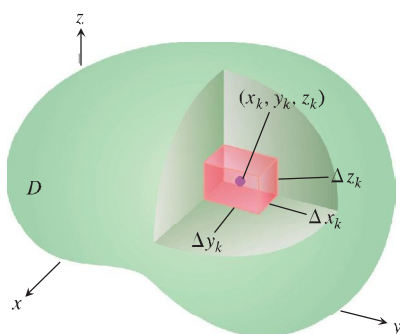


FIGURE 15.29 Partitioning a solid with rectangular cells of volume ΔV_k .